

SECTION 1 OVERVIEW

This manual includes hardware details and programming information for the MC68000, the MC68HC000, the MC68HC001, the MC68008, the MC68010, and the MC68EC000. For ease of reading, the name M68000 MPUs will be used when referring to all processors. Refer to M68000PM/AD, *M68000 Programmer's Reference Manual*, for detailed information on the MC68000 instruction set.

The six microprocessors are very similar. They all contain the following features

- 16 32-Bit Data and Address Registers
- 16-Mbyte Direct Addressing Range
- Program Counter
- 6 Powerful Instruction Types
- Operations on Five Main Data Types
- Memory-Mapped Input/Output (I/O)
- 14 Addressing Modes

The following processors contain additional features:

- MC68010
 - Virtual Memory/Machine Support
 - High-Performance Looping Instructions
- MC68HC001/MC68EC000
 - Statically Selectable 8- or 16-Bit Data Bus
- MC68HC000/MC68EC000/MC68HC001
 - Low-Power

All the processors are basically the same with the exception of the MC68008. The MC68008 differs from the others in that the data bus size is eight bits, and the address range is smaller. The MC68010 has a few additional instructions and instructions that operate differently than the corresponding instructions of the other devices.

1.1 MC68000

The MC68000 is the first implementation of the M68000 16-/32 bit microprocessor architecture. The MC68000 has a 16-bit data bus and 24-bit address bus while the full architecture provides for 32-bit address and data buses. It is completely code-compatible with the MC68008 8-bit data bus implementation of the M68000 and is upward code compatible with the MC68010 virtual extensions and the MC68020 32-bit implementation of the architecture. Any user-mode programs using the MC68000 instruction set will run unchanged on the MC68008, MC68010, MC68020, MC68030, and MC68040. This is possible because the user programming model is identical for all processors and the instruction sets are proper subsets of the complete architecture.

1.2 MC68008

The MC68008 is a member of the M68000 family of advanced microprocessors. This device allows the design of cost-effective systems using 8-bit data buses while providing the benefits of a 32-bit microprocessor architecture. The performance of the MC68008 is greater than any 8-bit microprocessor and superior to several 16-bit microprocessors.

The MC68008 is available as a 48-pin dual-in-line package (plastic or ceramic) and 52-pin plastic leaded chip carrier. The additional four pins of the 52-pin package allow for additional signals: A20, A21, $\overline{\text{BGACK}}$, and $\overline{\text{IPL2}}$. The 48-pin version supports a 20-bit address that provides a 1-Mbyte address space; the 52-pin version supports a 22-bit address that extends the address space to 4 Mbytes. The 48-pin MC68008 contains a simple two-wire arbitration circuit; the 52-pin MC68008 contains a full three-wire MC68000 bus arbitration control. Both versions are designed to work with daisy-chained networks, priority encoded networks, or a combination of these techniques.

A system implementation based on an 8-bit data bus reduces system cost in comparison to 16-bit systems due to a more effective use of components and byte-wide memories and peripherals. In addition, the nonmultiplexed address and data buses eliminate the need for external demultiplexers, further simplifying the system.

The large nonsegmented linear address space of the MC68008 allows large modular programs to be developed and executed efficiently. A large linear address space allows program segment sizes to be determined by the application rather than forcing the designer to adopt an arbitrary segment size without regard to the application's individual requirements.

1.3 MC68010

The MC68010 utilizes VLSI technology and is a fully implemented 16-bit microprocessor with 32-bit registers, a rich basic instruction set, and versatile addressing modes. The vector base register (VBR) allows the vector table to be dynamically relocated

1.4 MC68HC000

The primary benefit of the MC68HC000 is reduced power consumption. The device dissipates an order of magnitude less power than the HMOS MC68000.

The MC68HC000 is an implementation of the M68000 16-/32 bit microprocessor architecture. The MC68HC000 has a 16-bit data bus implementation of the MC68000 and is upward code-compatible with the MC68010 virtual extensions and the MC68020 32-bit implementation of the architecture.

1.5 MC68HC001

The MC68HC001 provides a functional extension to the MC68HC000 HCMOS 16-/32-bit microprocessor with the addition of statically selectable 8- or 16-bit data bus operation. The MC68HC001 is object-code compatible with the MC68HC000, and code written for the MC68HC001 can be migrated without modification to any member of the M68000 Family.

1.6 MC68EC000

The MC68EC000 is an economical high-performance embedded controller designed to suit the needs of the cost-sensitive embedded controller market. The HCMOS MC68EC000 has an internal 32-bit architecture that is supported by a statically selectable 8- or 16-bit data bus. This architecture provides a fast and efficient processing device that can satisfy the requirements of sophisticated applications based on high-level languages.

The MC68EC000 is object-code compatible with the MC68000, and code written for the MC68EC000 can be migrated without modification to any member of the M68000 Family.

The MC68EC000 brings the performance level of the M68000 Family to cost levels previously associated with 8-bit microprocessors. The MC68EC000 benefits from the rich M68000 instruction set and its related high code density with low memory bandwidth requirements.

